

Logarithmic Differentiation

Date_____ Period_____

Use logarithmic differentiation to differentiate each function with respect to x .

1) $y = 2x^{2x}$

2) $y = 5x^{5x}$

3) $y = 3x^{3x}$

4) $y = 4x^{x^4}$

5) $y = (3x^4 + 4)^3 \sqrt{5x^3 + 1}$

6) $y = (x^5 + 5)^2 \sqrt{2x^2 + 3}$

7) $y = \frac{(3x^4 - 2)^5}{(3x^3 + 4)^2}$

8) $y = \sqrt{3x^2 + 1} (3x^4 + 1)^3$